Social Media Engagement Analytics Using Python

Project Overview

The objective of this project was to analyze social media engagement data using Python and apply data analytics techniques such as data cleaning, transformation, exploratory data analysis (EDA), statistical analysis, visualization, and insight generation. The dataset contains information related to social media content performance, including likes, comments, shares, impressions, watch time, follower counts, user demographics, sentiment labels, and content categories. The analysis aims to identify factors influencing engagement, understand user behavior, evaluate content performance, and provide actionable business insights.

Task 1 – Data Import and Setup

The dataset was imported using the Pandas library and loaded into a DataFrame for analysis. Initial exploration was performed using:

head()	tail()	shape	columns	info()	dtypes

Date-related columns were converted into datetime format to facilitate time-series analysis and trend visualization.

Outcome

The dataset was successfully loaded and structured for further analysis. Data types were verified and corrected where necessary.

Task 2 – Data Cleaning and Transformation

Missing Value Handling

Missing values were identified using:

isnull()	isna()

Appropriate imputation techniques were applied:

- Mean/Median for numerical variables
- Mode for categorical variables
- Group-based imputation for engagement metrics where appropriate

Examples include:

- Age filled using median/mean values
- Gender filled using mode
- Sentiment labels filled using mode
- Likes, comments, and shares filled using group-level averages

Duplicate Handling

Duplicate records were identified using:

df.duplicated().sum()

No duplicate records were found in the dataset. Nevertheless, duplicate removal procedures were implemented to ensure data integrity.

Data Formatting

Data types were corrected and standardized.

Examples:

- User IDs converted to string/object type
- Post ID converted to string/object type
- Age converted to integer
- Numeric engagement metrics converted to proper numeric formats

Category Standardization

Categorical inconsistencies were cleaned.

Examples:

- Male, male, M standardized to Male
- Female, female, F standardized to Female
- Sentiment labels standardized to Positive, Neutral, and Negative

Feature Cleaning

A new feature called hashtag_count was created by extracting hashtags from post content. Sentiment labels were cleaned and standardized to improve consistency in downstream analysis.

Outcome

The dataset was successfully cleaned and transformed into a consistent, analysis-ready format.

Task 3 – Exploratory Data Analysis (EDA)

Several exploratory analyses were conducted to understand the dataset structure and identify patterns.

Dataset Overview

The following operations were performed:

head()	tail()	shape	columns	info()	dtypes

Categorical Analysis

Categorical variables were analyzed using:

value_counts()	unique()	nunique()

Examples include:

- Gender distribution
- Sentiment distribution
- Post type distribution
- Device type distribution

Correlation Analysis

A correlation matrix was generated for numerical variables to identify relationships among:

- Likes
- Comments
- Shares
- Watch Time
- Engagement Rate
- Follower Count

GroupBy Analysis

Aggregated summaries were created for:

- Post Type
- Country
- Sentiment

Metrics analyzed:

- Average Likes
- Average Comments
- Average Shares
- Average Engagement Rate
- Average Impressions

Outcome

EDA provided an understanding of user engagement behavior, content performance, and relationships among key metrics.

Task 4 – Data Wrangling

Several derived features were created to enhance analysis.

Engagement Score

A custom engagement score was created:

Engagement Score = Likes + Comments + Shares

This metric provided a unified measure of content performance.

Engagement Rate

Engagement Rate was analyzed to evaluate user interaction relative to audience size and reach.

Log Transformation

Log-transformed metrics were generated for selected variables to reduce skewness and improve interpretability.

Grouped Summaries

Detailed summaries were produced by:

- Post Type
- Country
- Sentiment

Outcome

New analytical features improved the ability to measure and compare content performance across different segments.

Task 5 – Statistical Analysis

Descriptive statistics were computed for:

- Likes
- Comments
- Shares
- Watch Time
- Engagement Rate
- Follower Count

Metrics calculated:

Mean	Median	Mode	Standard Deviation	Variance	Percentiles	Skewness	Kurtosis

Key Findings

Likes

The mean and median values were very close, indicating a balanced distribution of likes across posts.

Comments

Comment activity was distributed relatively evenly among posts.

Shares

Shares exhibited a stable distribution with limited influence from extreme outliers.

Watch Time

Watch time showed a broad range of user engagement but remained relatively symmetric.

Engagement Rate

Engagement rate displayed extremely high positive skewness and kurtosis, indicating that a small number of posts achieved viral performance and generated exceptionally high engagement.

Follower Count

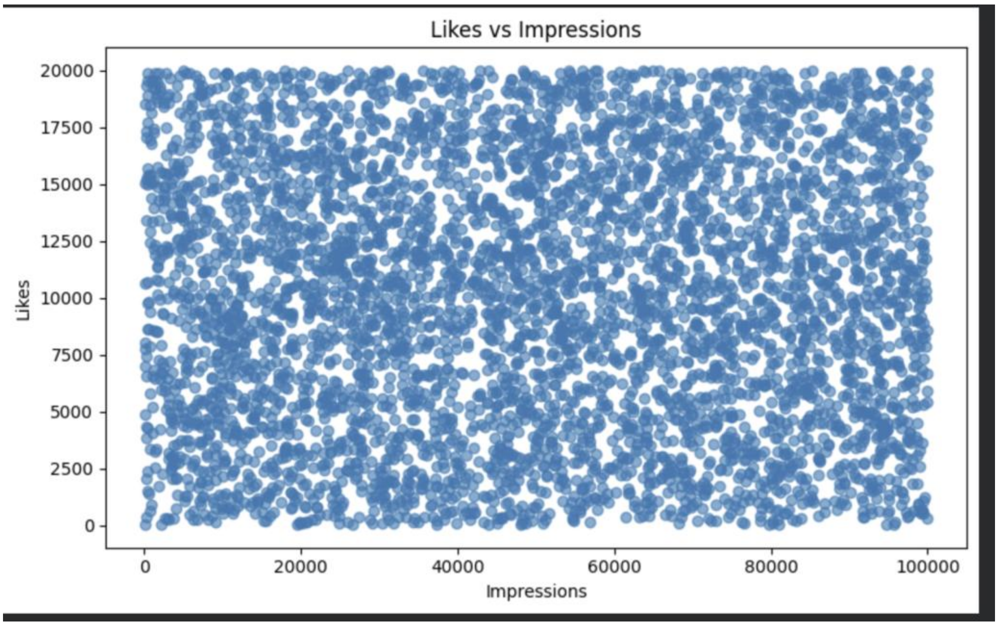
Follower counts were relatively balanced across users, with moderate variation between accounts.

Task 6 – Data Visualization

A total of more than sixteen visualizations were created using Matplotlib, Seaborn, and Plotly.

Matplotlib Visualizations

Scatter Plot: Likes vs Impressions



Observation

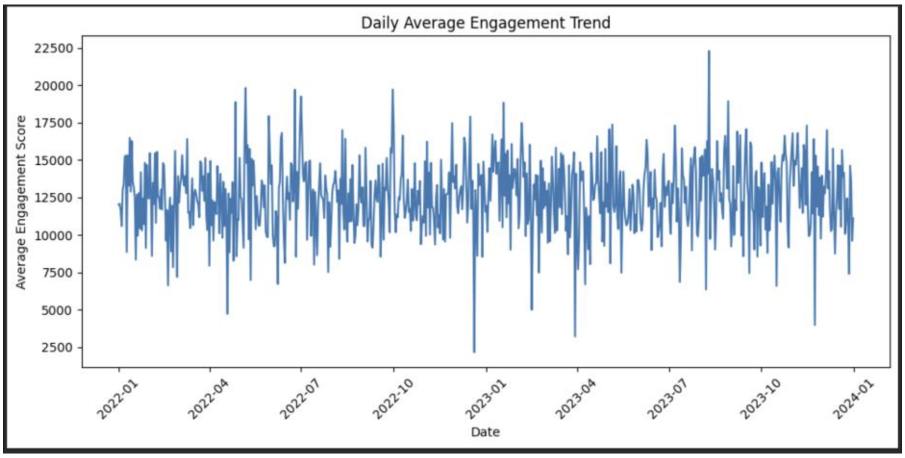
- The scatter plot shows a broad distribution of likes across all impression levels.
- Posts with both low and high impressions can receive either low or high numbers of likes.
- The points are widely dispersed, indicating a weak visual relationship between impressions and likes.

- No major clusters or distinct patterns are observed.

  Insight

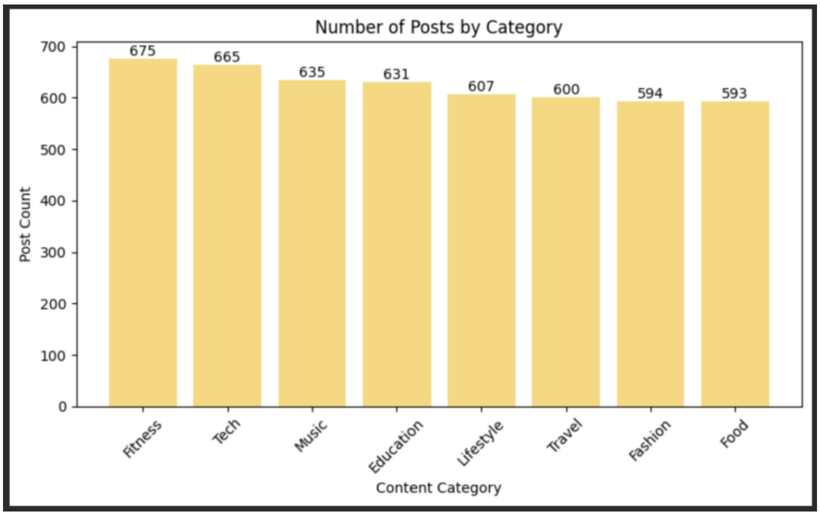
  The number of impressions alone does not appear to be a strong predictor of likes. While impressions increase content visibility, other factors such as content quality, relevance, sentiment, posting time, and audience characteristics likely play a significant role in determining user engagement.

Line Chart: Daily Engagement Trend



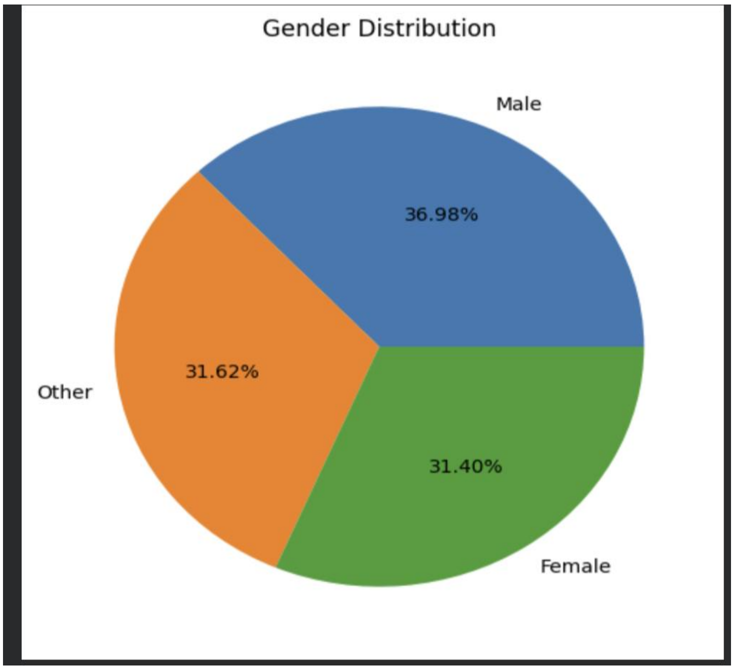
Aspect	Interpretation
Trend	No significant long-term increasing or decreasing trend is observed.
Stability	Average engagement remains relatively stable throughout the period.
Peaks	Several engagement spikes indicate highly successful content or campaigns.
Dips	Temporary drops suggest periods of lower audience interaction.
Business Insight	Consistent engagement levels indicate a stable audience base, while peak periods can be analyzed further to identify characteristics of high-performing content.

Bar Chart: Posts by Category



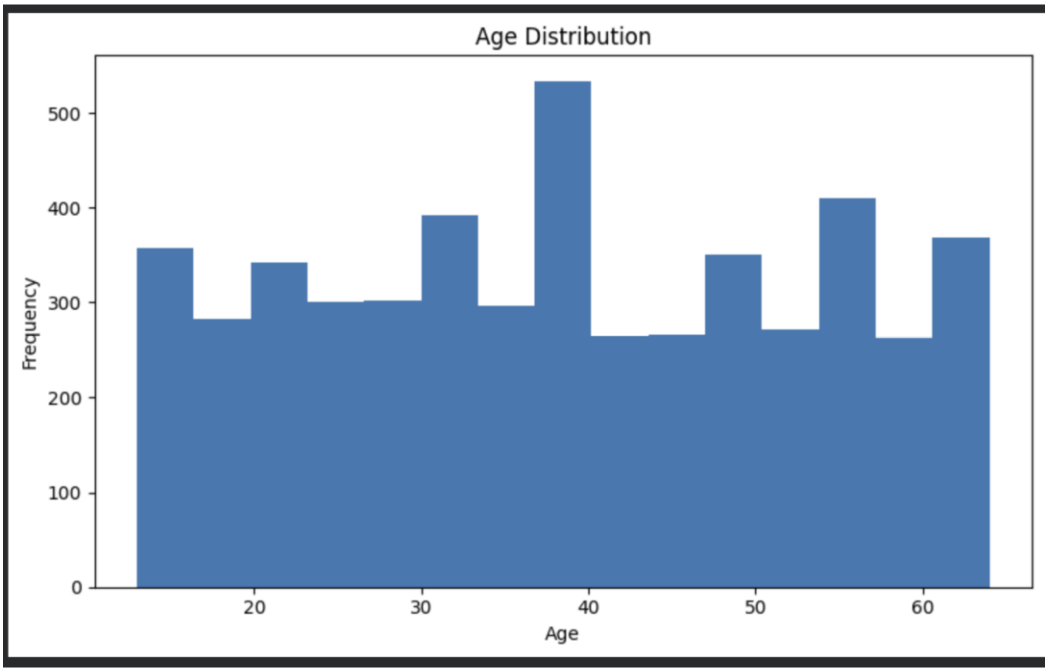
Aspect	Interpretation
Highest Category	Fitness has the highest number of posts (675).
Lowest Category	Food has the lowest number of posts (593).
Distribution	Categories are relatively evenly distributed.
Content Diversity	The platform supports a diverse range of content topics.
Business Insight	A balanced category distribution reduces content concentration risk and allows engagement analysis across multiple audience interests.

Pie Chart: Gender Distribution



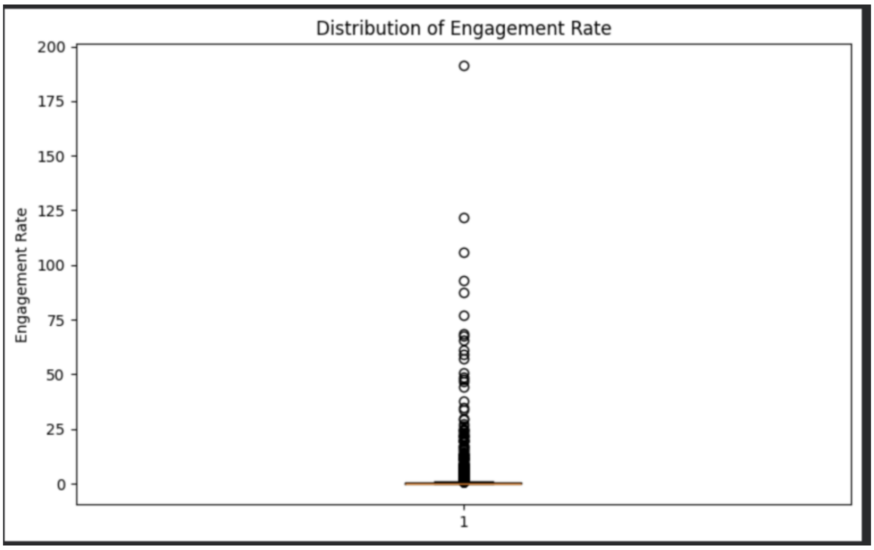
Aspect	Interpretation
Largest Segment	Male users constitute 36.98% of the dataset.
Female Representation	Female users account for 31.40% of users.
Other Category	Other users represent 31.62% of the dataset.
Distribution	Gender distribution is fairly balanced across all categories.
Diversity	The dataset demonstrates strong demographic diversity.
Business Insight	Since no single gender dominates the dataset, engagement patterns can be analyzed across genders with minimal representation bias.

Histogram: Age Distribution



Aspect	Interpretation
Age Range	Users range approximately from 13 to 65 years old.
Distribution	Ages are fairly evenly distributed across the dataset.
Peak Range	Slightly higher concentration is observed around ages 35–40.
Diversity	The dataset contains users from multiple age groups and generations.
Business Insight	The platform appeals to a broad audience, allowing content strategies to target multiple age segments rather than focusing on a single demographic.

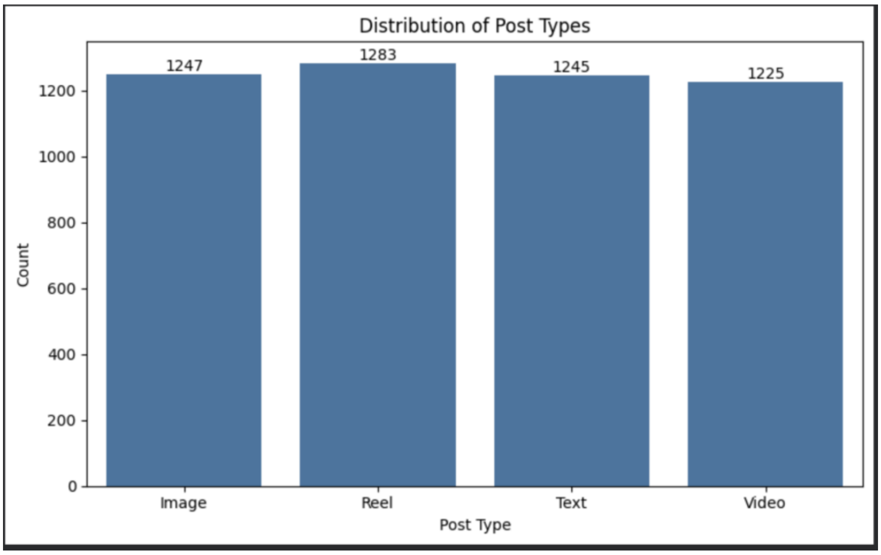
Box Plot: Engagement Rate Distribution



Aspect	Interpretation
Median	The median engagement rate is very low, indicating that most posts receive moderate engagement.
Spread	Most observations are concentrated within a narrow engagement range.
Outliers	Numerous extreme outliers are present above the upper whisker.
Distribution	The engagement rate distribution is highly positively skewed.
Business Insight	A small percentage of posts generate exceptionally high engagement, highlighting the importance of identifying and replicating characteristics of top-performing content.

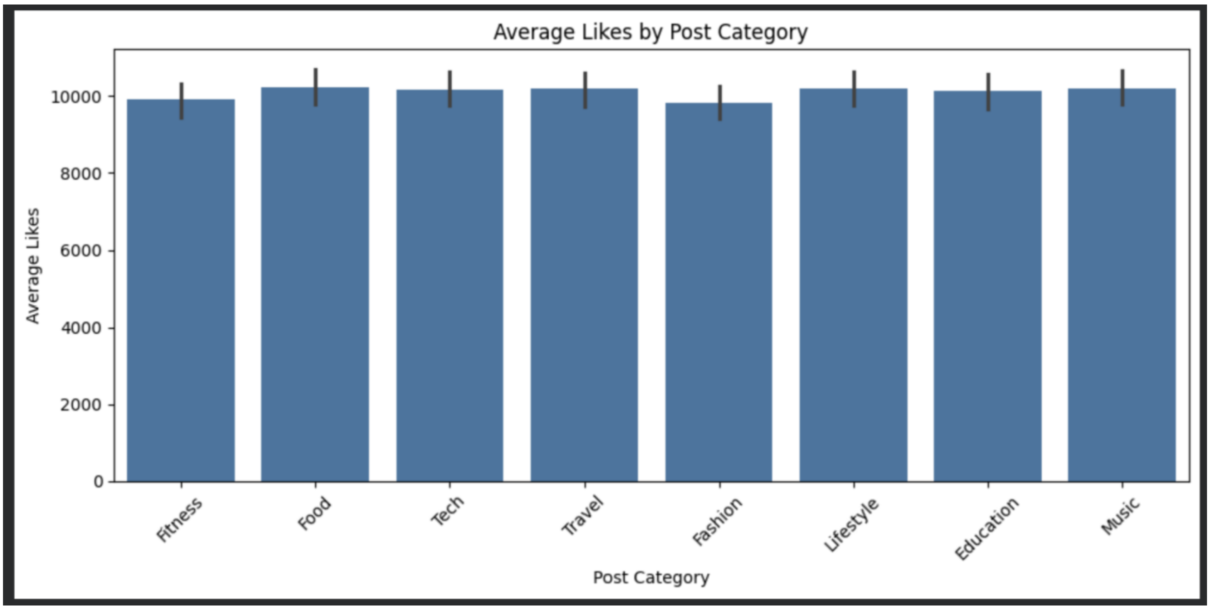
Seaborn Visualizations:

Count Plot: Post Type Distribution



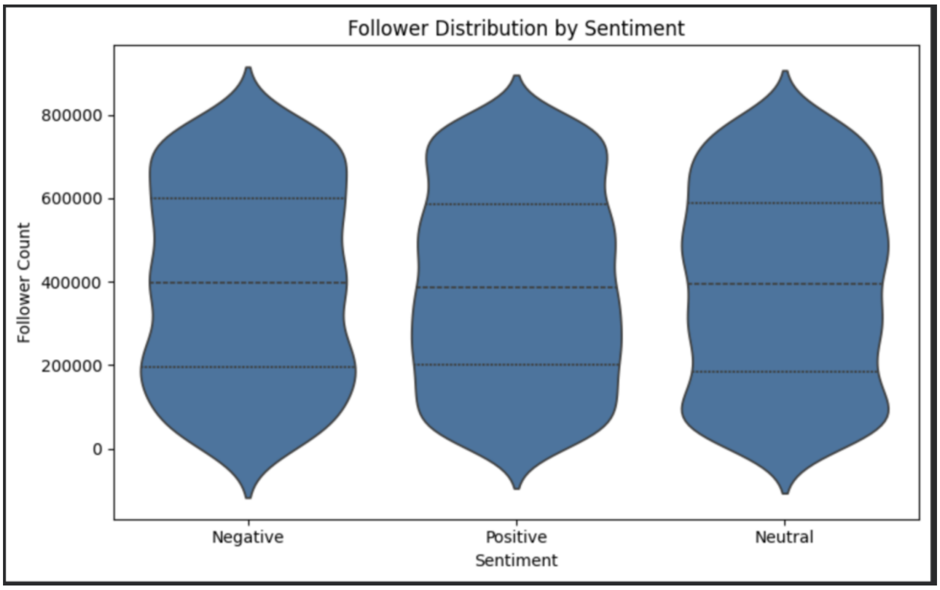
Aspect	Interpretation
Most Common Post Type	Reel (1,283 posts)
Least Common Post Type	Video (1,225 posts)
Distribution	All post types are nearly equally represented.
Content Strategy	The platform maintains a balanced mix of content formats.
Business Insight	Since post type frequencies are similar, differences in engagement are more likely due to content quality rather than posting frequency.

Bar Plot: Average Likes by Category



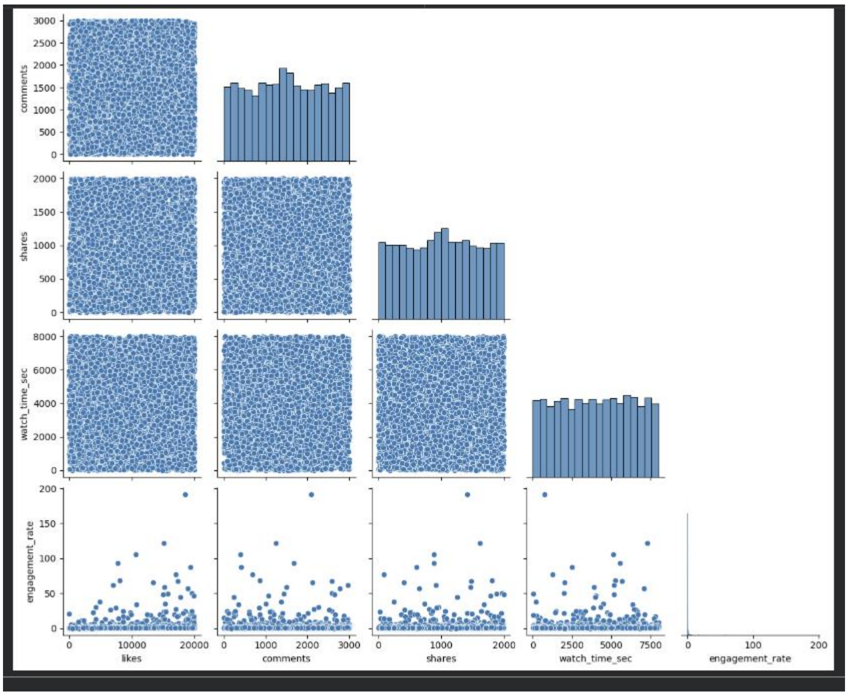
Aspect	Interpretation
Highest Average Likes	Food category receives the highest average likes.
Lowest Average Likes	Fashion category receives the lowest average likes.
Distribution	Average likes are very similar across all categories.
Variability	Error bars indicate moderate variation within categories.
Business Insight	Category selection alone does not significantly impact likes; content quality and audience relevance may play a larger role.

Violin Plot: Followers vs Sentiment



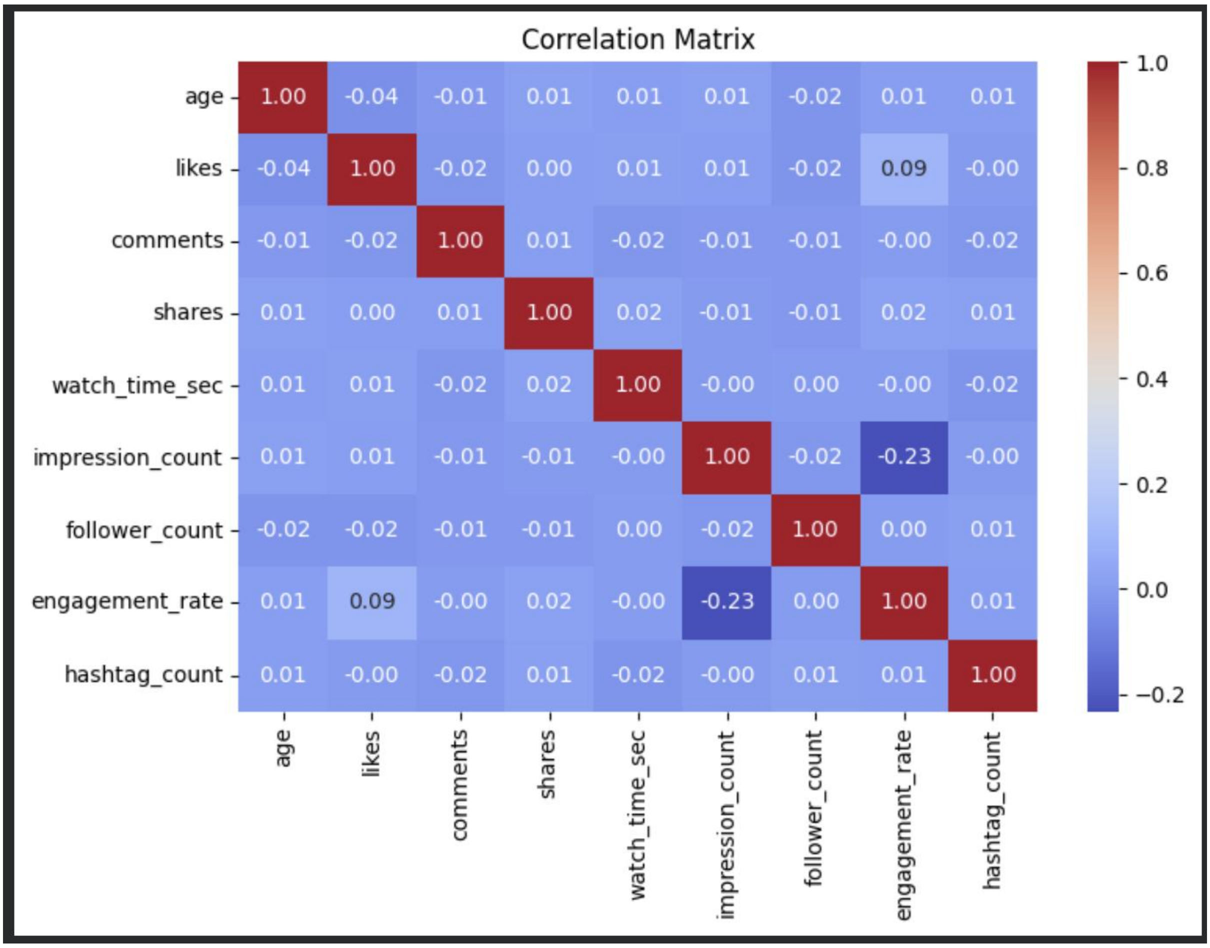
Aspect	Interpretation
Distribution	All sentiment categories show similar follower distributions.
Range	Follower counts span a broad range across all sentiments.
Median	Median follower counts are approximately the same for all groups.
Variation	Similar variability exists among Positive, Negative, and Neutral posts.
Business Insight	Sentiment does not appear to influence the follower size of content creators.

Pair Plot: Numeric Feature Relationships



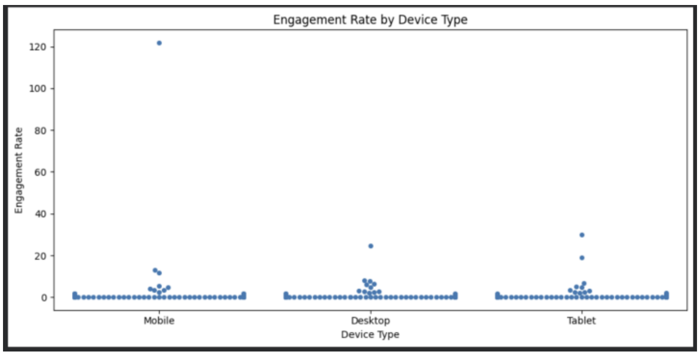
Aspect	Interpretation
Likes Distribution	Broadly distributed across the dataset.
Comments Distribution	Fairly uniform with no dominant range.
Shares Distribution	Spread across the full range of values.
Watch Time Distribution	Evenly distributed with moderate variability.
Engagement Rate	Highly skewed with numerous outliers.
Correlation	No strong visual relationships between variables.
Business Insight	User engagement appears to be driven by multiple factors rather than a single engagement metric.

Heatmap: Correlation Matrix



Finding	Correlation	Interpretation
Impression Count vs Engagement Rate	-0.23	Weak negative relationship
Likes vs Engagement Rate	0.09	Very weak positive relationship
Age vs Likes	-0.04	No meaningful relationship
Follower Count vs Engagement Rate	0.00	No relationship
Hashtag Count vs Engagement Metrics	≈0.00	No significant impact
Overall Matrix	Most values near 0	Variables are largely independent

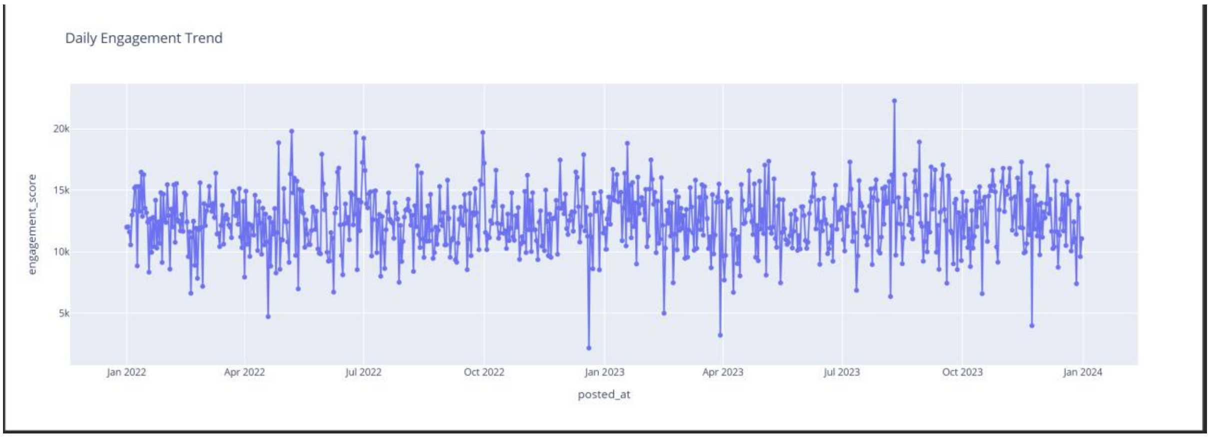
Swarm Plot: Engagement Rate vs Device Type



Aspect	Interpretation
Distribution	Most engagement rates are concentrated at low values across all devices.
Mobile	Contains the highest engagement-rate outlier (~122).
Desktop	Shows moderate variability with a few high-engagement posts.
Tablet	Includes some high-engagement outliers but fewer extreme values.
Overall Pattern	Device types exhibit broadly similar engagement distributions.
Business Insight	Device type does not appear to strongly influence engagement rate in this dataset.

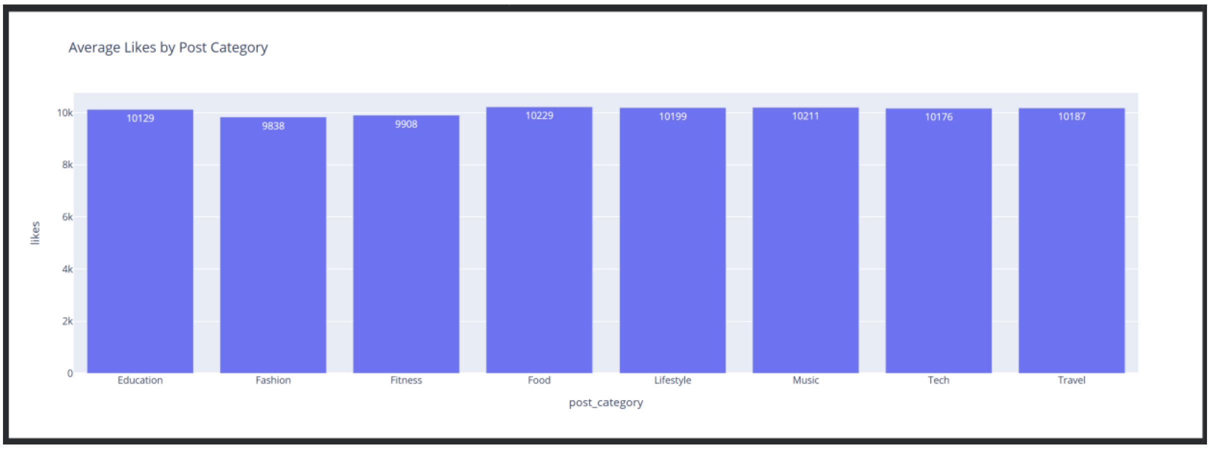
Plotly Interactive Visualizations

Interactive Line Chart



Aspect	Interpretation
Trend	No significant long-term increase or decrease is observed.
Stability	Engagement remains relatively stable throughout the period.
Peaks	Several days exhibit exceptionally high engagement (>20,000).
Dips	A few days experience unusually low engagement (<5,000).
Overall Pattern	Daily engagement fluctuates but remains centered around a consistent average level.
Business Insight	The platform maintains a stable audience, while certain posts or campaigns generate temporary engagement surges.

Interactive Bar Chart



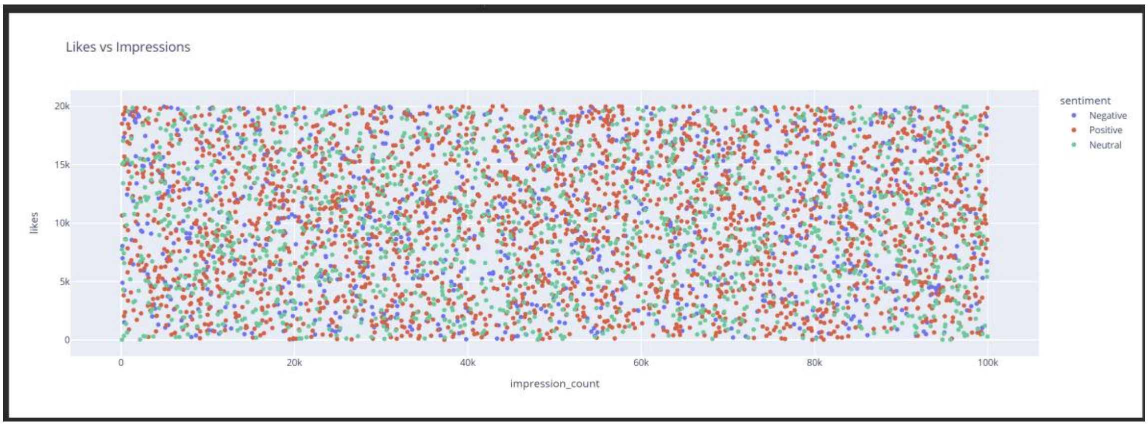
Key Findings

- Highest Performing Category: Food
- Lowest Performing Category: Fashion
- Overall Variation: Very small
- Audience Preference: Fairly balanced across categories

Business Insight

Since average likes are nearly identical across all content categories, creators should not rely solely on category selection to increase engagement. Other factors such as content quality, visual design, posting time, sentiment, hashtags, and audience targeting are likely to have a greater impact on performance.

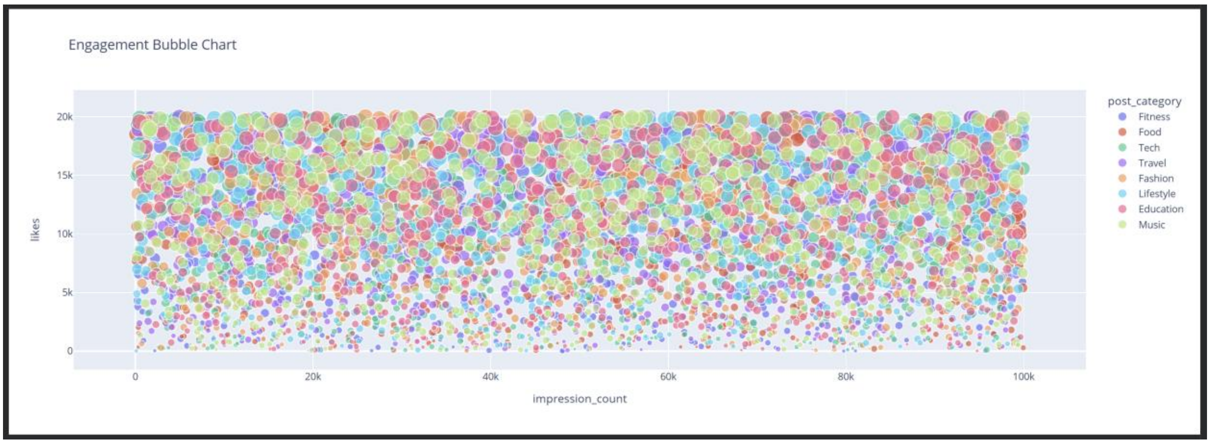
Interactive Scatter Plot



Aspect	Interpretation
Relationship	Weak relationship between impressions and likes.
Sentiment Impact	Positive, Neutral, and Negative posts show similar patterns.
Clustering	No distinct clusters are visible.

Aspect	Interpretation
Trend	No clear positive or negative correlation.
Business Insight	High reach does not automatically guarantee high engagement.

Interactive Bubble Chart

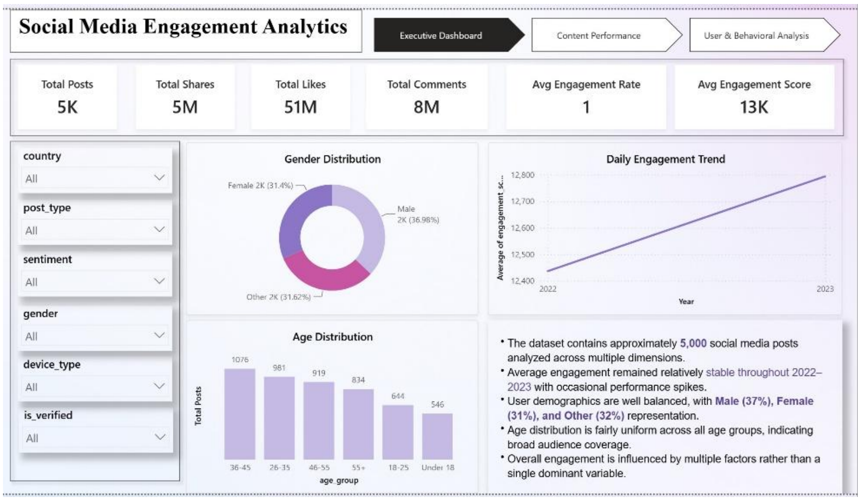


Aspect	Interpretation
Impressions vs Likes	Weak relationship observed.
Bubble Size	Larger bubbles indicate higher engagement scores.
High Engagement Posts	Mostly associated with higher likes.
Category Performance	High-performing posts exist across all categories.
Business Insight	Content quality influences engagement more than impressions alone.

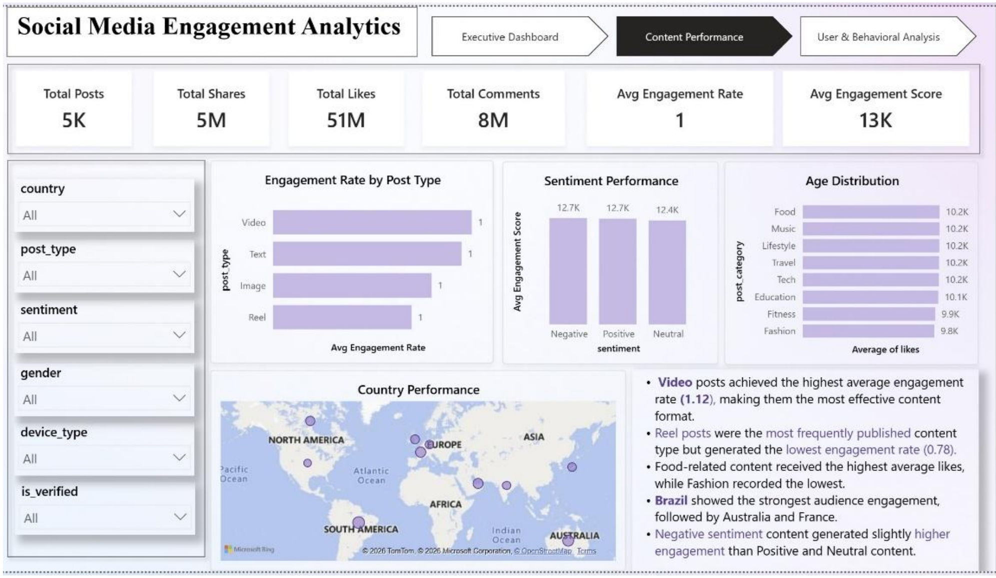
Each visualization was accompanied by observations and business interpretations.

Dashboard in Power BI and Python Script:

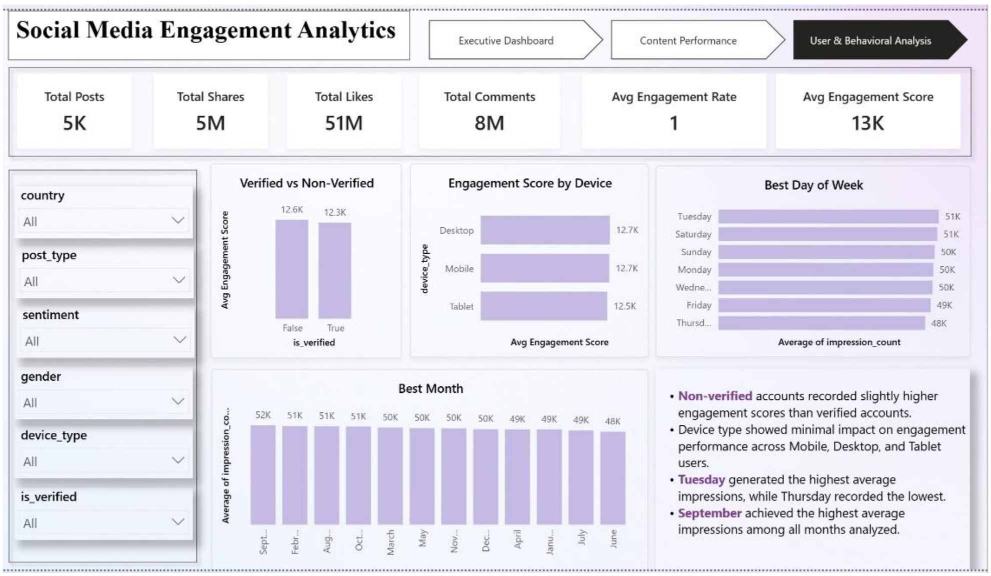
Executive Dashboard



Content Performance



User & Behavioral Analysis



Final Insights:

1. Content Performance

Post Type Performance

1. Video posts generated the highest average engagement rate (1.12), making them the most effective content format for audience interaction.
2. Text posts ranked second (1.06) and also performed strongly.
3. Image posts achieved moderate engagement (0.90).
4. Reels recorded the lowest average engagement rate (0.78) despite being the most frequently published post type.

5. This suggests that increasing the quantity of a content format does not necessarily improve engagement performance.

Business Insight:

1. Video content appears to be the most effective format for driving audience interaction.
2. Content creators may benefit from prioritizing high-quality video content over simply increasing Reel volume.

Country Performance

1. Brazil achieved the highest average engagement rate (1.54) among all countries.
2. Australia and France also demonstrated strong audience engagement.
3. USA recorded the lowest average engagement rate (0.58) despite being one of the largest digital markets.
4. Engagement rates varied considerably across countries, indicating geographic differences in audience behavior.

Business Insight:

1. Audience engagement is not uniform across regions.
2. Content strategies that perform well in Brazil, Australia, and France may not produce the same results in the USA or India.
3. Organizations should consider local audience preferences and regional content strategies when planning social media campaigns.

Posting Date Analysis

Due to the absence of time-of-day information, posting behavior was analyzed using calendar dates. The analysis identified the highest-performing days/months in terms of average impressions.

1. September achieved the highest average impressions (51,514), while June recorded the lowest (48,134).
2. Tuesday generated the highest average impressions (51,309), followed by Saturday and Sunday.
3. Thursday produced the lowest average impressions (48,119).
4. Monthly and weekly variations were relatively small, indicating stable audience visibility throughout the year.

Business Insight:

1. Although certain months and days slightly outperform others, the differences are modest.

2. Consistently producing relevant and engaging content is likely more important than relying solely on posting schedules.

Category Performance

1. Food posts achieved the highest average likes (10,229), while Fashion posts recorded the lowest (9,838).
2. However, differences across categories were relatively small.
3. This suggests that content category alone was not a major driver of engagement.

Reach vs Engagement

1. The Likes vs Impressions scatter plot showed a weak relationship between impressions and likes.
2. High impressions did not necessarily result in high engagement.
3. Content quality and audience relevance appear to be more important than visibility alone.

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2. User Trends

Age and Engagement

1. Correlation analysis showed almost no relationship between age and engagement metrics.
2. User engagement remained relatively consistent across different age groups.
3. Age was not a strong predictor of content performance.

Verified vs Non-Verified Accounts

Key Findings

1. Non-verified accounts achieved a higher average engagement score (12,645.49) compared to verified accounts (12,309.47).
2. The difference between the two groups is approximately 336 engagement points.
3. Contrary to expectations, verified accounts did not generate the highest engagement in this dataset.

Business Insight

Audience engagement appears to depend more on content quality, relevance, and audience interest than on verification status. While verified accounts may have greater credibility or visibility, non-verified creators can still achieve comparable or even higher engagement through compelling content.

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3. Behavioral Insights

Engagement Trends Over Time

1. Daily engagement remained stable throughout the analysis period (2022–2024).
2. Several short-term engagement spikes were observed, indicating occasional high-performing content.
3. No long-term upward or downward trend was identified.

Device Type Impact

1. Mobile, Desktop, and Tablet users exhibited similar engagement-rate distributions.
2. Mobile contained the highest engagement outlier, but overall device differences were minimal.
3. Device type did not appear to strongly influence engagement performance.

Watch Time Behavior

1. Watch time showed weak correlations with other engagement metrics.
2. Higher watch time alone did not guarantee higher likes, shares, or engagement rates.

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4. Sentiment Analysis

Key Findings

1. Negative sentiment posts generated the highest average engagement score (12,732.34).
2. Positive sentiment posts ranked second (12,668.92).
3. Neutral sentiment posts recorded the lowest average engagement score (12,446.04).
4. The difference between the highest and lowest sentiment category is approximately 286 points, which is relatively small compared to the overall engagement levels.

Business Insight

Content containing stronger emotional signals, particularly negative sentiment, tends to generate slightly higher audience interaction. However, the small gap between Negative and Positive sentiment indicates that both emotionally engaging positive and negative content can perform well. Neutral content generally attracts lower engagement, suggesting that emotional relevance may be an important factor in driving audience participation.

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5. Correlation Analysis

Key Findings

1. Most correlation coefficients were close to zero.
2. The strongest observed relationship was a weak negative correlation between Impression Count and Engagement Rate (-0.23).

3. Likes showed only a weak positive relationship with Engagement Rate (0.09).

Interpretation

No single variable strongly explained engagement performance. Content success appears to be influenced by multiple interacting factors rather than one dominant metric.

Conclusion

The project successfully applied Python-based data analytics techniques to analyze social media engagement data. Data cleaning, transformation, exploratory analysis, statistical analysis, and visualization were performed to uncover meaningful insights.

While content type and sentiment showed some influence on engagement, most variables exhibited weak correlations with engagement metrics. This suggests that content performance is driven by a combination of factors rather than a single dominant predictor. While most engagement metrics exhibit stable distributions, engagement rate is heavily influenced by a small number of high-performing posts, demonstrating the significant impact of viral content. The findings from this analysis can help organizations optimize content strategies, improve audience engagement, and make data-driven decisions for future social media campaigns.